

Omar Farooq

AI/ML Engineer & Data Scientist

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Professional Summary

AI/ML Engineer with 1 year of hands-on industry experience building end-to-end, production-oriented AI systems across speech recognition, agentic/LLM workflows, computer vision, MLOps, and developer tooling. Currently contributing to government-scale AI initiatives at PITB. Experienced in dataset creation, model fine-tuning, evaluation, MLflow-based experiment tracking, annotation workflows, API/backend development, and deployment on Linux/Docker environments. Previously worked in data analytics and automation, translating analysis into measurable business impact.

Experience

Associate AI/ML Engineer | Punjab Information Technology Board (PITB)

Feb 2026 - present

- Iteratively **fine-tuned Whisper Small, Medium, and Large V3** across multiple experiment cycles, diagnosing training/configuration issues and driving **WER** down from an early **90% to 13.61%** through systematic dataset cleaning, deduplication, and scaling; **explored LoRA/quantized training to stabilize the 1.5B-parameter Large model under GPU memory constraints**.
- Created and expanded custom Urdu speech datasets (Common Voice, FLEURS, custom YouTube podcast/interview audio), **growing the merged corpus beyond 150 hours**, **Lead a team of Data Annotators to curate own Datasets** including building a **YouTube audio-extraction and AI-assisted transcription pipeline for original data collection**.
- Built a **YOLO + Label Studio + MLflow human-in-the-loop pipeline** covering model-assisted annotation, immutable exports, dataset versioning, scheduled retraining, and model promotion gates.
- Built & Deployed **LDA/PHATA compliance/document-intelligence system** using **OCR (qwen), GptOSS, RAG, structured SOP rule extraction**, evidence retrieval, rule-level evaluation, human review, and audit-ready outputs.
- Lead 60 + Interns** for the summer batch at PITB.

Junior Data Analyst | Game District

Aug 2025 - Feb 2026

- Automated ad-network performance analysis with Python**, reducing a recurring reporting workflow from roughly **4 hours to under 5 seconds**.
- Built an end-to-end Power BI monetization pipeline using API-sourced data and performed EDA across 350+ datasets spanning 15+ ad networks**.
- Designed and analyzed **70+ A/B tests** and contributed to monetization improvements, including a reported **150% AdMob revenue rise**; **recognized as Employee of the Month**.

Selected Projects & Engineering Work

- SQL Safety Proxy v1.0.0 ([GitHub](#))**: Developer-focused database safety proxy with fail-closed SQL policy enforcement, impact estimation, prepared-statement inspection, transaction recovery, audit logging, Docker runtime, 341 automated tests, and real PostgreSQL E2E validation.
- Urdu Automatic Speech Recognition (Organizational [GitLab](#))**: End-to-end Urdu speech engineering program spanning custom dataset creation, annotation leadership, Whisper fine-tuning, MLflow tracking, and benchmarking.
- YOLO Human-in-the-Loop Pipeline (Organizational [GitLab](#))**: Continuous-learning object detection workflow integrating Label Studio, YOLO, immutable annotation exports, dataset lifecycle management, retraining, and promotion controls.
- LDA/PHATA Compliance Evaluator (Organizational [GitLab](#))**: Document-intelligence pipeline using OCR, RAG, structured SOP rules, evidence-grounded compliance evaluation, human review, and automated reporting.

Technical Skills

- AI / ML**: Python, PyTorch, TensorFlow, Scikit-learn, Hugging Face, Whisper, YOLO, OpenCV
- LLM / Agentic**: LangGraph, LlamaIndex, RAG, MCP, Ollama, structured outputs, prompt engineering, tool-using agents
- MLOps / Infra**: MLflow, Docker, Linux, CUDA, GitHub Actions, GitLab, experiment tracking, model evaluation
- Backend / Data**: FastAPI, Flask, PostgreSQL, SQL, REST APIs
- Analytics**: Pandas, NumPy, Power BI, Matplotlib, Excel, exploratory data analysis, A/B testing

Certifications

- AI Engineer Agentic Track: The Complete Agent & MCP Course** | Udemy (Ed Donner / Ligncy) Aug 2026
[Certificate Link](#)
- Data Science & Machine Learning using Python** | Lahore University of Management Sciences (LUMS) May 2025
[Certificate Link](#)

Education

Bachelors Degree Software Engineering | COMSATS University Islamabad, Lahore Campus

Sep 2021 - Jul 2025

Additional

Technical writing: Published content on practical data/ML topics on Medium - [Read Here](#)